INCH - POUND

MIL-C-47069B(MI)
5 June 1991
SUPERSEDING
MIL-C-47069A(MI)
30 September 1985

MILITARY SPECIFICATION

COATING, ELECTRICALLY CONDUCTIVE, AIR-DRYING

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of coating, electrically conductive, air-drying.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8030

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SPECIFICATIONS

FEDERAL

L-P-383	-	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated
P-P-101	-	Paper, Abrasive, Silicone Carbide, Waterproof
TT-T-548	-	Toluene, Technical
PPP-C-2020	-	Chemicals, Liquid, Dry and Paste: Packaging of

MILITARY

DOD-P-15328	-	Primer (Wash), Pretreatment (Formula No. 117 for Metals) (Metric)

STANDARDS

FEDERAL

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing
FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 4615	-	n-Butyl Acetate (All Grades), Standard Specification for

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. The coating shall be composed of 40 to 50 percent by volume of silver powder suspended in a butyl acetate vehicle with a suitable resin binder.

3.3 Physical and electrical properties.

3.3.1 Condition in container. A freshly opened full container of coating shall show no hard or gummy cake and shall mix readily to a smooth homogeneous state.

3.3.2 Storage stability. The coating, after storage for not less than one year in an unopened container, shall show no evidence of curdling, livering, or caking, and shall be free from grit, lumps, and skins (see 4.6.2.2).

3.3.3 Dilution stability. The coating shall show no signs of separation or curdling when thinned with butyl acetate, conforming to ASTM D 4615, to produce a viscosity of 80 to 100 centipoises (cP) (see 4.6.2.3).

3.3.4 Spraying properties. A sprayed coating of 0.001 to 0.002 inches shall dry to a uniform, smooth surface free from running, sagging, or streaking (see 4.6.2.4).

3.3.5 Adhesion. The coating shall show no signs of separation from the primed surface to which it is applied when tested (see 4.6.2.5).

3.3.6 Drying time. The applied coating shall dry hard within 24 hours in air at a temperature of 23.9 $\pm$ 2.8° Celsius (C) and a relative humidity of 50 $\pm$ 4 percent (see 4.6.2.6).

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3.3.7 Surface resistivity. The applied coating shall have a surface resistivity which shall be not greater than 3.0 ohms per square (see 4.6.2.7).

3.3.8 Heat resistance. The applied coating shall show no evidence of cracking, blistering, or peeling when tested as specified (see 4.6.2.8).

3.3.9 Humidity resistance. The applied coating shall show no evidence of cracking or peeling after being subjected to the humidity test (see 4.6.2.9).

3.4 Workmanship. The coating material shall be uniform in quality and free from dirt and other foreign material.

3.5 Material safety data sheets. A material safety data sheet shall be prepared in accordance with the requirements of FED-STD-313 (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.6.1.

4.4 First article inspection. When specified in the contract or order (see 6.2), a first article sample shall be subjected to first article inspection. Subsequent units shall not be considered for acceptance, until Government approval of the first article sample has been obtained.

4.5 Inspection provisions.

4.5.1 Lot size. A lot shall consist of material produced from the same batch of blending operations from one manufacturer and one unchanged process. In the event of a continuous process, a lot shall consist of material subjected to the same processing operations and conditions.

4.5.2 Sampling.

4.5.2.1 For visual examination. Unless otherwise specified (see 6.2), sampling for visual examination shall be in accordance with table I.

TABLE I. Sampling plan A.

Lot size	Sample size
2 to 5	All
6 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.5.2.2 For examination testing. Unless otherwise specified (see 6.2), sampling for examination testing shall be in accordance with table II.

4.5.2.3 For inspection testing. Unless otherwise specified (see 6.2), sampling for inspection testing shall be in accordance with table II.

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TABLE II. Sampling plan B.

Lot size	Sample size
2 to 3	All
4 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.5.3 Examination.

4.5.3.1 Visual examination. Visual examination of the samples specified in 4.5.2.1 shall be performed to determine compliance with the requirements specified in 3.2 and 3.4, and preparation for delivery specified in section 5.

4.5.3.2 Examination testing. Examination testing of the samples specified in 4.5.2.2, to determine compliance with the following characteristics, shall be performed in accordance with their corresponding test paragraphs in accordance with table III.

TABLE III. Characteristics of examination testing.

Characteristic	Test
Condition in container (see 3.3.1)	4.6.2.1
Dilution stability (see 3.3.3)	4.6.2.3
Spraying properties (see 3.3.4)	4.6.2.4
Adhesion (see 3.3.5)	4.6.2.5

4.5.4 Inspection testing. Inspection testing of the sample specified in 4.5.2.3, to determine compliance with the following characteristics, shall be performed in accordance with their corresponding test paragraphs in accordance with table IV.

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TABLE IV. Characteristics of inspection testing.

Characteristic	Test
Storage stability (see 3.3.2)	4.6.2.2
Drying time (see 3.3.6)	4.6.2.6
Surface resistivity (see 3.3.7)	4.6.2.7
Heat resistance (see 3.3.8)	4.6.2.8
Humidity resistance (see 3.3.9)	4.6.2.9

4.5.5 Inspection equipment. The Government shall determine whether the contractor has available, and uses correctly, gaging, measuring, and test equipment of required accuracy and precision. The instruments shall be a proper type and range to make measurements within the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments to conduct regularly scheduled calibrations of his inspection equipment. Records of calibrations shall be maintained by the contractor and made available for Government review. The calibration of gages, standards, and instruments shall be checked periodically by Government personnel.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified, tests shall be performed at 23.9 ± 2.8°C.

4.6.1.1 Preparation of test panels. Two sets of test panels shall be prepared. One set shall be on aluminum and prepared in accordance with Method 2013.1 of FED-STD-141. Unless otherwise specified, 2024 aluminum alloy shall be used. They shall be given one coat pretreatment primer as specified in DOD-P-15328 and allowed to air-dry for not less than half an hour. The other set of test panels shall be of 3-inch by 0.125 inch sheets of laminated polyester-glass fiber conforming to L-P-383. The panels shall be sanded smooth with abrasive paper, grit size 320, as specified in P-P-101, and wiped clean with toluene conforming to TT-T-548. The panels shall then be given one coat of pretreatment primer as specified in DOD-P-15328 and allowed to air-dry for not less than half an hour.

4.6.2 Tests. Tests requiring the use of test panels shall be performed using each of the two types of tests panels specified in 4.6.1.1, except for 4.6.2.7 which shall be performed using only the laminated polyester-glass fiber test panel.

4.6.2.1 Condition in container. A freshly opened container of the coating material shall be tested in accordance with Method 3011.2 of FED-STD-141 to determine compliance with 3.3.1.

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4.6.2.2 Storage stability. The coating material shall be tested in accordance with Method 3022.1 of FED-STD-141 to determine compliance with 3.3.2.

4.6.2.3 Dilution stability. The coating material, after being thinned as specified in 3.3.3, shall be tested in accordance with Method 4203.1 of FED-STD-141.

4.6.2.4 Spraying properties. The coating material, after being thinned as specified in 3.3.3, shall be tested to determine compliance with 3.3.4 in accordance with Method 4331.1 of FED-STD-141 with the following exceptions:

a. The test panel shall be prepared and primed as specified in 4.6.1.1.

b. The primed test panel shall be sprayed with the coating material to obtain a dry film thickness of 0.001 to 0.002 inch and air-dried for 24 hours.

4.6.2.5 Adhesion. The coating material, after being applied to a test panel as specified in 4.6.2.3, shall be tested, to determine compliance with 3.3.5, in accordance with Method 6301.2 of FED-STD-141, except there shall be no scratching of the specimen.

4.6.2.6 Drying time. A test panel, sprayed and air-dried as specified in 4.6.2.3, shall be tested in air having a relative humidity of 50 $\pm$ 4 percent for a dry-hard finish in accordance with Method 4061.2 of FED-STD-141 to determine compliance with 3.3.6.

4.6.2.7 Surface resistivity. Electrodes shall be mounted on a test panel, sprayed and air-dried as specified in 4.6.2.3, so that each electrode forms opposite sides of a square which shall be not less than 0.25 inch. Firm contact with the coating shall be made by applying pressure to the electrodes. With a potential of 3 $\pm$ 2 volts direct current applied across the electrodes, the resistance between the electrodes shall be measured with an ohmmeter to determine compliance with 3.3.7.

4.6.2.8 Heat resistance. A test panel, sprayed and air-dried as specified in 4.6.2.4, shall be tested, mounted in a horizontal position, in accordance with Method 6051 of FED-STD-141 for 2 hours at 71 $\pm$ 1.1°C. At the end of the 2 hour period, the coating on the test panel shall be examined to determine compliance with 3.3.8.

4.6.2.9 Humidity resistance. A test panel sprayed and air-dried as specified in 4.6.2.4, shall be placed in a humidity chamber and subjected to aggravated humidity cycling to determine compliance with 3.3.9. The test shall be conducted in accordance with MIL-STD-810, Method 507.3, Procedure III, for a duration of 240 hours.

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4.7 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of PPP-C-2020.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing and marking of the coating material shall be in accordance with PPP-C-2020 (see 6.2). In addition, each unit shall be marked with its lot number and date of manufacture.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use as an air-drying, heat resistant, electrically conductive coating over metal and plastic. A primer coat is necessary to ensure good adhesion under humid conditions.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

c. When first article sample is required (see 3.1 and 4.4)

d. Methods of sampling and inspection, if other than specified (see 4.5.2.1, 4.5.2.2, and 4.5.2.3)

e. When packaging and packing for overseas or domestic shipping is required (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the

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bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 Subject term (keyword) listing.

Adhesive
Conductor
Insulator

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Civilian Coordinating Activity:
GSA - FSS

Preparing Activity:
Army - MI

(Project No. 8030-A126)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-47069B(MI)	910605

3. DOCUMENT TITLE
COATING, ELECTRICALLY CONDUCTIVE, AIR-DRYING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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